

MSC BUSINESS ANALYTICS

TEXT ANALYTICS GROUP PROJECT

TOTAL MARKS [50marks]

QUESTION

Text Analytics is a process of extracting useful patterns, relationships, or insights from text data. It involves using various techniques from natural language processing (NLP), machine learning, and data mining to analyze and transform unstructured text into structured data.

Each group will undertake one of the assigned **Text Analytics case projects** in the table below. The objective is to apply Natural Language Processing, machine learning and data visualisation techniques to a real-world textual dataset and develop an evidence-based solution to the assigned problem.

Structure

- Data cleaning; **Regular Expression**
- Text preprocessing
- Exploratory text analytics
- Text representation: **Word embeddings**
- Building models: Model 1 and 2
- Model evaluation: **Recall, Precision etc...**
- Model comparison
- Best model
- Interpretation
- Use Python with Streamlit

Final Written Report

Each group should submit a **6 -10-page technical report**, excluding references and appendices.

- Cover Page
- Background
- Approach (Methodology)
- Results
- Interpretation

Also prepare a **presentation decks (up to 10pages)** to present the project in class. Be prepared to demonstrate your work to the class.

Deploy the project on free sever and share the link: Streamlit Community Cloud

Deadline: 3rd September 2026

Link:

Presentation Date: TBD

2. Group Project Allocation

Group	Case Project	Model Evaluation	Embeddings / Representation	Algorithms
1	Customer Review Sentiment Analysis	• Accuracy, • Precision, • Recall, F1, • Confusion Matrix, • ROC-AUC	TF-IDF	Logistic Regression + SVM
2	Emotion Detection	• Accuracy, • Precision, • Recall, F1, • Confusion Matrix, • ROC-AUC	BERT	SVM + Logistic Regression
3	News Topic Classification	• Accuracy, • Precision, • Recall, F1, • Confusion Matrix, • ROC-AUC	Word2Vec	Naïve Bayes + SVM
4	Restaurant/Recipe Review Analytics	• Accuracy, • Precision, • Recall, F1, • Confusion Matrix, • ROC-AUC	TF-IDF	SVM + Random Forest
5	Movie Review Sentiment	• Accuracy, • Precision, • Recall, F1, • Confusion Matrix, • ROC-AUC	GloVe	Naïve Bayes + LSTM
6	SMS Spam Detection	• Accuracy,	TF-IDF	Naïve Bayes + SVM

		• Precision, • Recall, F1, • Confusion Matrix, • ROC-AUC		
7	Fake News Detection	• Accuracy, • Precision, • Recall, F1, • Confusion Matrix, • ROC-AUC	BERT	Logistic Regression + SVM
8	Customer Complaint Topic Modelling	• Topic coherence, • topic diversity, • topic interpretability, • topic distribution	TF-IDF	LDA + NMF
9	Social Media Sentiment	• Accuracy, • Precision, • Recall, F1, • Confusion Matrix, ROC-AUC	TF-IDF	SVM + Logistic Regression
10	Aspect-Based Sentiment Analysis	• Accuracy, • Precision, • Recall, F1, • Confusion Matrix, ROC-AUC	GloVe	SVM + Logistic Regression
11	Toxic Comment Detection	• Accuracy, • Precision, • Recall, F1, • Confusion Matrix, • ROC-AUC	TF-IDF + BERT	Logistic Regression + SVM

12	Student Feedback Analytics	• Accuracy, • Precision, • Recall, F1, • Confusion Matrix, • ROC-AUC	TF-IDF + Word2Vec	SVM + Random Forest
13	Llama-Based Customer-Service Chatbot — No RAG	• Answer correctness, • relevance, helpfulness, • fluency, hallucination/error rate	LLaMA token/embedding representation	LLaMA + supervised fine-tuning/instruction tuning
14	News Topic Modelling	• Topic coherence, • topic diversity, • topic interpretability	TF-IDF	LDA + NMF
15	Multilingual Sentiment Analysis	• Accuracy, • Precision, • Recall, F1, • Confusion Matrix, • ROC-AUC	FastText	SVM + Logistic Regression
16	NLP Model Benchmarking	• Accuracy, • Precision, • Recall, F1, • Confusion Matrix, • ROC-AUC	TF-IDF	SVM + Logistic Regression
17	Llama-Based Customer-Service Chatbot — No RAG	• Answer correctness, • relevance, helpfulness, • fluency, hallucination/error rate	LLaMA token/embedding representation	LLaMA + supervised fine-tuning/instruction tuning

3. DATASET

Use the dataset assigned to your project. **Only use the dataset that corresponds to your group number.**

Group	Suggested Data Source
1	https://archive.ics.uci.edu/dataset/331/sentiment+labelled+sentences
2	https://huggingface.co/datasets/dair-ai/emotion
3	https://scikit-learn.org/stable/datasets/real_world.html
4	https://archive.ics.uci.edu/dataset/911/re
5	https://ai.stanford.edu/~amaas/data/sentiment/
6	https://archive.ics.uci.edu/dataset/228/sms+spam+collection
7	https://www.kaggle.com/datasets/aadyasingh55/fake-news-classification
8	https://www.consumerfinance.gov/data-research/consumer-complaints/
9	https://huggingface.co/datasets/cardiffnlp/tweet_sentiment_multilingual
10	https://alt.qcri.org/semeval2014/task4/
11	https://www.kaggle.com/competitions/jigsaw-toxic-comment-classification-challenge
12	https://www.kaggle.com/datasets/ziya07/contextual-sentiment-feedback-dataset
13	https://huggingface.co/datasets/bitext/Bitext-customer-support-llm-chatbot-training-dataset
14	https://scikit-learn.org/stable/datasets/real_world.html
15	https://huggingface.co/datasets/mteb/multilingual-sentiment-classification
16	https://archive.ics.uci.edu/dataset/331/sentiment+labelled+sentences
17	https://huggingface.co/datasets/bitext/Bitext-customer-support-llm-chatbot-training-dataset